

UNIVERSITY OF CALIFORNIA, DAVIS

Department of Statistics

3D & 2D Visualization of Wildfire Risk in Glendale, CA

STA 160 — Team 3 Capstone Project (Visualization-Heavy)

Project Team:

<i>Name</i>	<i>Major</i>	<i>Role</i>
Alexander Davis (Lead)	Statistics	GIS Lead Dev, 3D Scene, 2D Leaflet Map, Website Build
Ruhi Aggarwal	Data Science	Building Risk Model
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1. Abstract

This project evaluates wildfire risk in Glendale, California through GIS-based terrain modeling, vegetation classification, and building-level exposure scoring. Using a USGS digital elevation model, WHR13 vegetation rasters, and LARIAC building footprints, we derived slope, aspect, and fuel-type layers and reclassified them onto a standardized 1–5 risk scale. A weighted composite model—50% vegetation, 30% slope, and 20% aspect—was constructed to map the spatial distribution of wildfire susceptibility across the city. Building-level risk categories were appended using logic-based rules derived from terrain and exposure metrics. The final outputs include an interactive 3D visualization built with Qgis2threejs and a 2D Leaflet dashboard for querying structure-level risk. Results show that the steep chaparral slopes along Glendale’s eastern and northern hillsides correspond to the highest wildfire exposure, highlighting the importance of terrain–fuel interactions in local hazard planning.

2. Introduction & Motivation

Wildfire hazard is not only a statewide concern but a particularly acute issue in the Los Angeles region, where steep topography, dense chaparral vegetation, and expanding wildland–urban interface (WUI) zones create a high-exposure environment. Glendale has experienced repeated nearby fire activity—including the Station Fire (2009), La Tuna Fire (2017), Eaton Fire (2025) and multiple Verdugo Mountain ignitions—which collectively highlight the vulnerability of its hillside neighborhoods. These events have repeatedly strained emergency response capacity and underscored the influence of terrain-driven fire behavior on evacuation time, flame spread, and structural damage.

Despite the availability of statewide Fire Hazard Severity Zone (FHSZ) maps, these products operate at coarse regional scales and do not provide building-level resolution needed for localized planning. Glendale’s unique combination of steep slopes, south-facing aspects, and highly flammable chaparral fuels means that risk varies significantly over short spatial distances—differences that are not captured in broad statewide hazard classifications.

Our project addresses this gap by producing a high-resolution, building-level wildfire risk visualization that integrates terrain, vegetation, and structural characteristics. By using open geospatial data, transparent methods, and interactive deployment, we aim to provide a tool that can support neighborhood-level hazard awareness, guide mitigation planning, and illustrate how environmental factors interact to create fire-prone micro-regions within Glendale. This level of spatial granularity is essential for emergency preparation, community outreach, and long-term resilience in a region where wildfire remains a persistent and evolving threat.

3. Data and Methods

3.1 Data Sources

We used four categories of publicly available spatial datasets:

- **Elevation:** USGS 1/3-arc-second DEM (~10 m resolution). Basis for slope and aspect derivation.
- **Vegetation:** California WHR13 vegetation raster (California Natural Resources Agency). Contains fuel-type codes for chaparral, forest, grassland, agriculture, and urban areas.
- **Buildings:** LARIAC6 Building Footprints (2020). Polygon outlines and height attributes.
- **Administrative Boundaries:** Official City of Glendale boundary shapefile.

All datasets were processed using **QGIS 3.42**, the **Processing Toolbox**, and **Qgis2threejs** for 3D export. Deployment relied on **Leaflet.js**, **Plotly**, and **Three.js**.

3.2 Preprocessing Workflow

To ensure consistency and spatial alignment, all datasets were standardized:

- **Reprojection:** All geospatial layers were reprojected into a common coordinate system (EPSG:3857) to ensure accurate alignment. The building footprint dataset, originally stored in EPSG:6424, required conversion before any spatial operations could be performed.
- **Clipping:** Raster and vector layers clipped to the Glendale boundary.
- **Raster Preparation:** The elevation-based layers (slope, aspect) and vegetation data were imported as TIFF rasters. Each raster was clipped to the Glendale city boundary to reduce processing overhead and focus on the study area.
- **Raster Cleaning:** Raster values of **-9999** (NoData) were removed or masked out. Vegetation rasters containing WHR13 codes were preserved because they were necessary for fuel type classification.
- **Vector Cleaning:** Buildings with height=0 or NULL were removed for the 3D scene (due to processing limits with Qgis2threejs). All buildings were kept for the 2D map in Leaflet.
- **Building Footprint Preparation:** The raw building dataset was cleaned prior to integration:
 - Redundant building types (e.g., *Courtyard*) were removed.
 - Missing HEIGHT values were filled using measured statistics.
 - Geometry validity was checked and repaired to avoid processing failures.

This unified geospatial dataset formed the foundation for zonal statistics and model-ready feature extraction.

3.3 Terrain Derivation: Slope and Aspect

Slope and aspect were generated from the DEM:

- **Slope:** Degrees of steepness, 0°–90°
- **Aspect:** Downslope compass direction, 0–360°

These represent key physical drivers of fire spread and drying rates.

3.4 Fire-Risk Reclassification

To convert the raw slope, aspect, and vegetation layers into comparable fire-risk scores, each raster was reclassified onto a standardized **1–5 scale**, where 1 indicates *low risk* and 5 indicates *extreme risk*.

3.4.1 Slope Reclassification

Steeper slopes accelerate fire spread; therefore, slopes above 20° received the highest risk score.

<i>Slope (°)</i>	<i>0–2</i>	<i>2–5</i>	<i>5–10</i>	<i>10–20</i>	<i>20–90</i>
<i>Risk Score</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>

3.4.2 Aspect Reclassification

Fire risk from the aspect reflects sun exposure. South- and southwest-facing slopes receive more sunlight, drying vegetation and increasing flammability.

<i>Aspect (°)</i>	<i>0–45</i>	<i>45–135</i>	<i>135–225</i>	<i>225–315</i>	<i>315–360</i>
<i>Direction</i>	<i>N–NE</i>	<i>E–SE</i>	<i>S–SW</i>	<i>W–NW</i>	<i>N</i>
<i>Risk Score</i>	<i>1</i>	<i>3</i>	<i>5</i>	<i>3</i>	<i>1</i>

3.4.3 Vegetation Reclassification

Vegetation types were assigned fire-risk scores based on their fuel load and flammability. WHR13 codes were grouped into the following classes:

Vegetation Type (Code)	Risk	Impact / Explanation
Urban / Barren (URBXX)	1	Minimal fuel
Agriculture (AGCXX, CROP)	2	Managed low vegetation
Grassland (GRSLND)	3	Burns quickly but with low fuel
Forest (FRSTXX)	4	High biomass, moderate-high risk
Chapparral / Shrubland (SHRBLND, CHAPPXX)	5	Highly flammable, dominant in SoCal

3.5. Composite Risk Score

The three reclassified rasters were combined using a weighted linear model:

$$- \text{Composite Risk} = 0.3 * \text{Slope Risk} + 0.2 * \text{Aspect Risk} + 0.5 * \text{Vegetation Risk}$$

Weighting vegetation at 50% reflects its dominant influence on fire behavior in Southern California's chaparral ecosystems. The output was saved as *fire_risk_final.tif*.

3.6. Raster-to-Building Integration via Zonal Statistics

Our primary analytical objective was to attach environmental risk factors to each building. The workflow consisted of:

1. **Raster Alignment:** All rasters were clipped to the Glendale boundary and resampled as necessary.
2. **Polygon-Based Extraction:** Using Zonal Statistics, we computed:
 - o Mean slope (degrees)
 - o Mean aspect (0–360 orientation)
 - o Majority vegetation code (“fuel type”)
3. **Centroid Generation:** Centroid coordinates allowed linking with statistical models and enabled spatial indexing.
4. **Merging for Final Dataset:** After zonal statistics, the building table was merged with the fire-risk table (logic_based_clustering.csv) via building ID.

3.7 Building Fire-Risk Scoring

To assign fire-risk values to individual structures, we evaluated two complementary, non-supervised methods:

3.7.1. Rule-Based Scoring (*Deterministic Model*)

A simple, interpretable, domain-driven model was created using slope, aspect, and vegetation type.

Each building inherited the terrain values beneath its footprint and was assigned a 1–5 risk score based on fire-science principles (e.g., steep slopes and south-facing aspects increase fire spread potential) and weights for each of the three features: 40% slope, 30% aspect, 30% vegetation.

This approach does not use training labels and relies entirely on expert-informed thresholds.

3.7.2. Unsupervised Clustering (*K-Means with 3 Features*)

To explore whether the data naturally grouped buildings into meaningful categories, we applied:

- The elbow method to understand how many groupings may naturally occur with only 3 features
- K-Means clustering with $k = 5$ to identify groupings of potential fire risk using slope, aspect, and vegetation data

Clusters were then interpreted as relative “risk categories” based on their average slope, aspect, and vegetation characteristics.

3.7.3. Unsupervised Clustering (*PCA + K-Means*)

To explore whether the data naturally grouped buildings into meaningful categories, we applied:

- PCA to take into account all features (including building shape/size in addition to slope, aspect, and vegetation) while reducing dimensionality, and emphasize dominant structural/terrain patterns
- K-Means clustering on the new values to see if other building characteristics reveal any additional information

3.7.3. Comparison of Methods

Although neither approach is truly “supervised” (we lack historical fire-damage labels), comparing them is still informative:

- The logic-based and K-Means with 3 features methods agreed for ~89% of buildings, indicating consistent patterns in the underlying terrain and fuel structure.
- Disagreements highlight neighborhoods where risk is more complex or ambiguous.

This comparison helps validate the robustness of the rule-based system and identifies areas where additional data (e.g., historical fire perimeters, structure loss) would be required for a fully supervised model.

Together, these methods provide:

- A transparent rule-based baseline
- A data-driven perspective on how buildings cluster based on their physical environment
- Insight into whether risk categories are stable across modeling approaches

3.8 Visualization & Deployment

To communicate spatial fire-risk patterns intuitively, we developed a 3D model using Qgis2threejs.

1. **Building Extrusion:** Extrusion height was mapped directly from cleaned HEIGHT fields.
2. **Terrain Rendering:** DEM was converted into a 3D mesh to visualize slopes, ridges, and canyons.
3. **Risk-Based Styling:** Buildings were color-coded using a graduated ramp based on fire-risk score or cluster ID, helping highlight high-risk zones.
4. **Interactivity:** Although Qgis2threejs had limits, the exported model allowed clicking on buildings to view embedded attributes, offering transparency and interpretability.

Outputs were deployed through a fully interactive website with:

- A **3D terrain + buildings viewer** (Qgis2threejs)
- A **2D Leaflet map** for filtering buildings by risk
- A **Plotly dashboard** containing drought trends and historical wildfire severity plots.

4. Results and Interpretation

The final composite terrain-based fire-risk raster shows high-risk zones clustered along the Verdugo Mountain foothills, where steep chaparral-covered slopes intersect with south- and southwest-facing aspects. Building-level results reinforce this pattern. Structures located on the eastern and northeastern boundary of Glendale consistently appear in the “Very High” and “Extreme” categories.

Comparison between the rule-based building-risk model and the K-Means clustering approach demonstrated strong alignment (88.8% agreement across 54,089 buildings). This suggests that both human-interpretable fire-science rules and unsupervised data-driven techniques identify similar classes of structurally exposed areas. Buildings where the two methods disagree are primarily those located along steep transitional terrain or irregular vegetation-fuel boundaries, indicating areas that may warrant closer field investigation or higher-resolution data. The PCA analysis resulted in clustering into 2 different categories, and given the average values of each group, we can conclude that one group is likely residential buildings while the other is likely commercial buildings.

The 2D dashboard visualizations provide historical context:

- Wildfire frequency has risen across Los Angeles County and the Angeles National Forest in the last decade.
- Decadal acreage burned shows substantial spikes during drought-intensive periods.
- A top-down building-risk map shows that the highest structural risks occur along narrow canyons, steep ridgelines, and the wildland–urban interface.

Collectively, these results highlight Glendale as a community with substantial exposure to topographically driven wildfire hazards.

5. Discussion and Reflection

The final visualization effectively highlights fire risk relative to urban density, but the process surfaced several core challenges in geospatial data science—especially data harmonization and 3D rendering.

Our first issue was data heterogeneity. The DEM and fire risk rasters came from different sources with different Coordinate Reference Systems (CRS) and units, causing misalignment. We resolved this by re-projecting all layers into a single projected CRS and standardizing units so slope and elevation calculations were consistent. This reinforced that unnoticed CRS mismatches can cascade into major analytical errors, making metadata verification an essential first step.

We also had to determine how to assign continuous environmental data to discrete building polygons. Because buildings may span multiple raster cells, we evaluated zonal statistics and centroid methods, ultimately aggregating the underlying raster values to compute mean slope, aspect, and vegetation for each building. This provided a clear, building-level risk profile across Glendale.

Data anomalies posed another challenge: many buildings listed a height of zero. For accurate 3D rendering, we removed these invalid records to avoid flat artifacts that would undermine the visualization. However, for the 2D Leaflet map, we kept all buildings to preserve overall city completeness—highlighting the common trade-off between data quality and model performance.

Finally, constructing the 3D map required integrating realistic terrain and building models. Using the DEM for a terrain mesh and extruding buildings by their cleaned height values allowed the visualization to clearly convey how Glendale’s steep, vegetation-dense slopes interact with the built environment. This 3D perspective proved critical for communicating the spatial reality of fire risk.

6. Acknowledgement and References

The following data providers and tools were utilized:

- U.S. Geological Survey. (2023). *USGS 1/3 arc-second Digital Elevation Model (DEM)*. <https://www.usgs.gov/>
- Cal Fire. (2023). *WHR13 Vegetation Dataset*. <https://www.fire.ca.gov/>
- Los Angeles County GIS Data Portal. (2020). *LARIAC6 Building Footprints*. <https://egis-lacounty.hub.arcgis.com/>
- City of Glendale GIS Division. (2023). *City Boundary Shapefiles*. <https://www.glendaleca.gov/>
- California Department of Forestry and Fire Protection (CAL FIRE). (2023). *FRAP Historical Wildfire Perimeters & Fire Hazard Severity Zones*. <https://frap.fire.ca.gov/>
- Mapbox. (2025). *Basemap Tiles*. <https://www.mapbox.com/>
- Graham, Robert. “CVA Drought Risk.” *County of Los Angeles Open Data*, 30 July 2024, data.lacounty.gov/datasets/lacounty::cva-drought-risk/about
- California Department of Forestry and Fire Protection. (n.d.). *Fire and Resource Assessment Program*. <https://www.fire.ca.gov/what-we-do/fire-resource-assessment-program>

Software and tools used include QGIS 3.42, Qgis2threejs, Leaflet.js, Plotly, and GitHub.

AI Assistance:

LLM’s were used for code debugging, modeling risk scores for buildings, editing written content, HTML/CSS improvements, and refinement of the report narrative. All analytical decisions, modeling approaches for terrain fire-risk, and visualizations were developed by the project team.

APPENDIX:

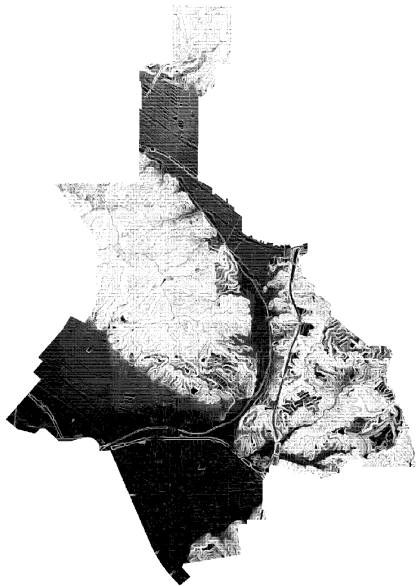
This appendix provides supplementary plots, code snippets, screenshots, and implementation details referenced in the main report. Materials here support the primary analysis.

A. Supplementary Figures

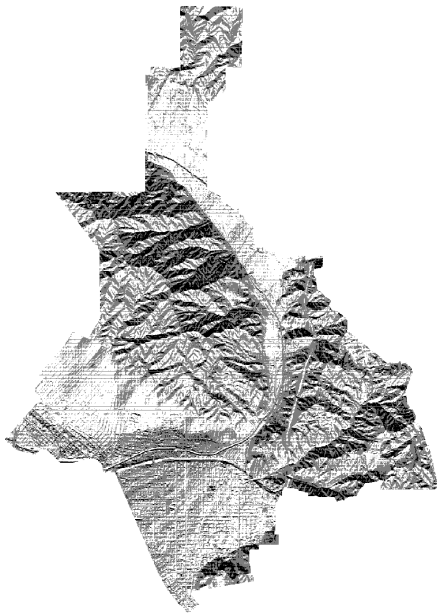
A.1 Terrain-Derived Risk Layers

Note: lighter colors in the raster layer corresponded to higher risk scores

- **Slope (degrees)** – classified into 1–5 risk bins



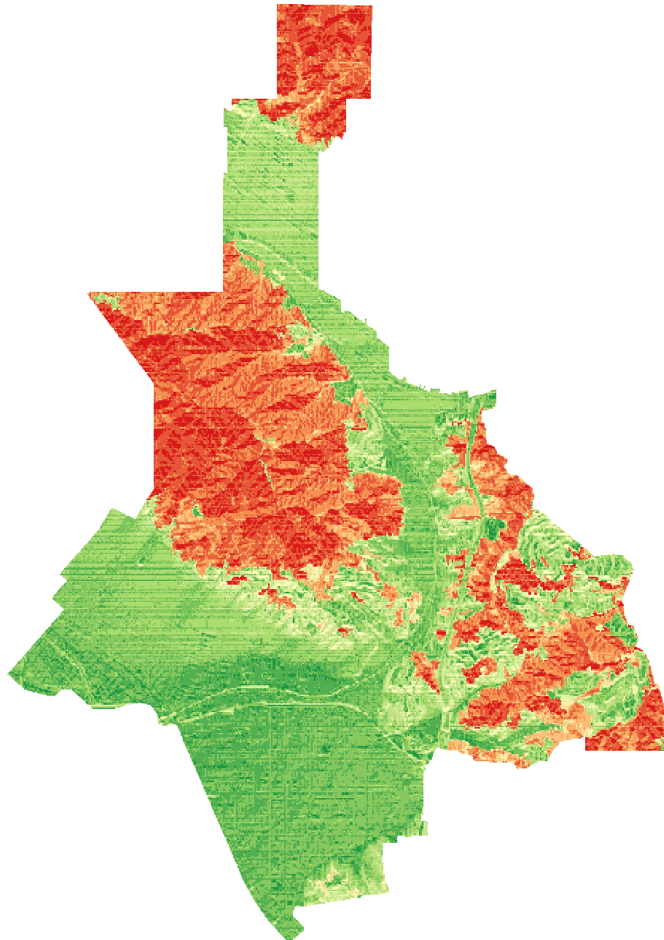
- **Aspect (azimuth)** – classified into cardinal aspect fire-risk categories



- **Vegetation (WHR13)** – categorized by fuel type

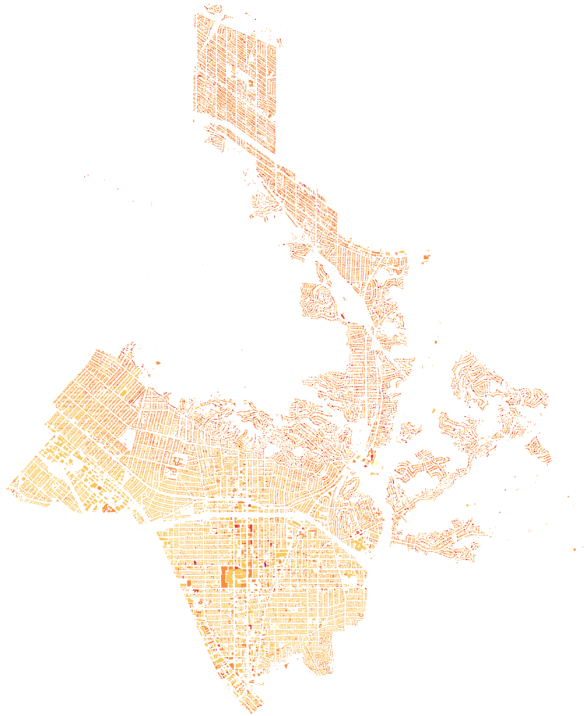


- These layers are combined to form a composite fire risk map with an inverse RdYlGr color ramp:



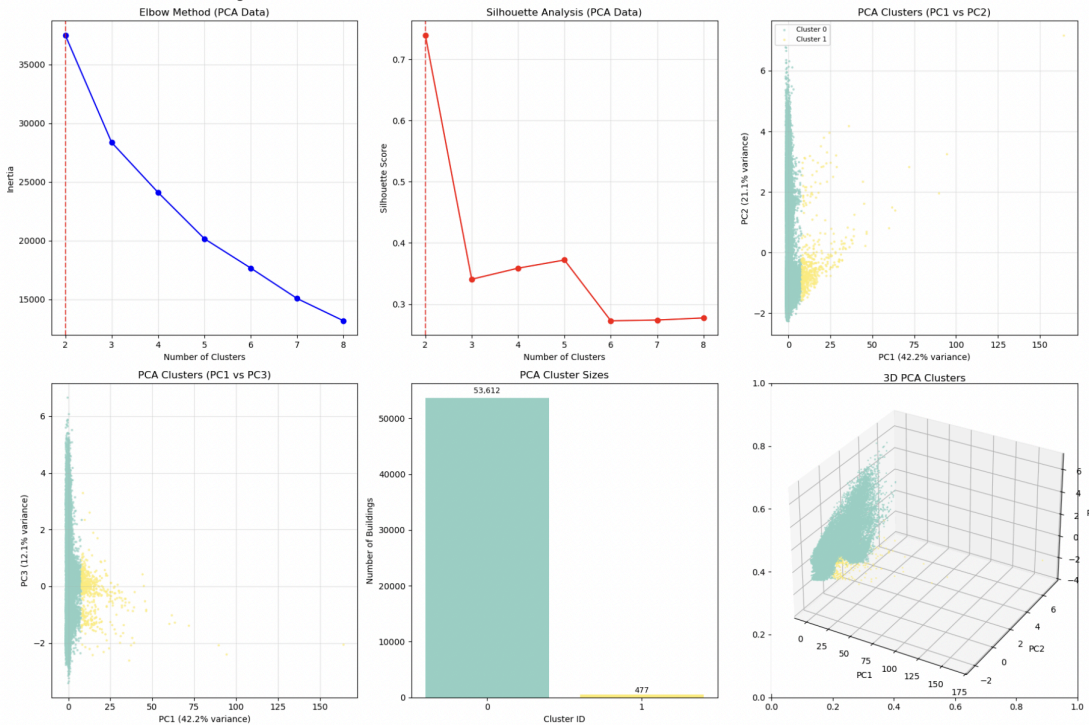
A.2 Building-Level Risk Outputs

- Building footprints with rule-based scores assigned to a YlOrRd color ramp

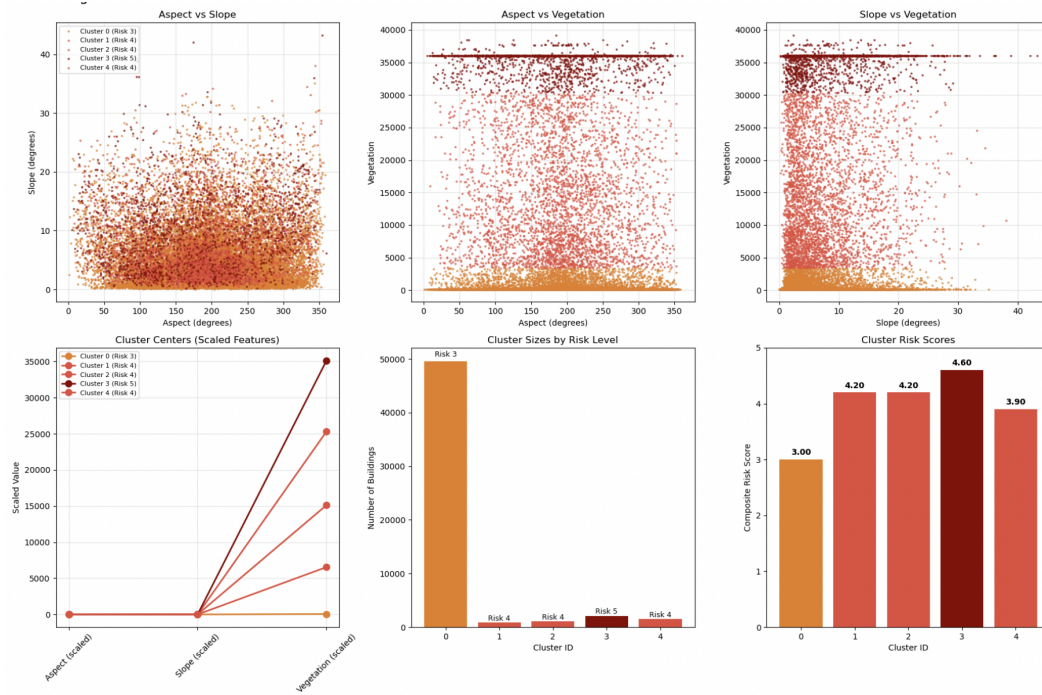


- PCA cluster distributions (buildings)

PCA Cluster distribution:
Cluster 0: 53,612 buildings (99.1%)
Cluster 1: 477 buildings (0.9%)



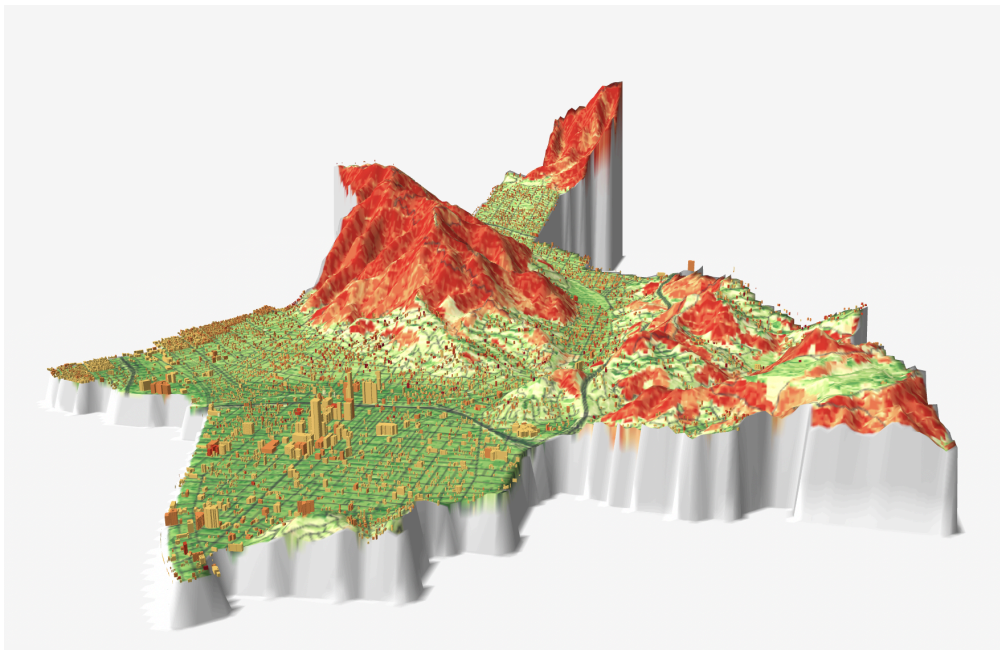
- Cluster visualizations (buildings)



A.3 Interactive Web Tool Screenshots

3D Viewer (Qgis2threejs)

- Initial view of the terrain and extruded buildings

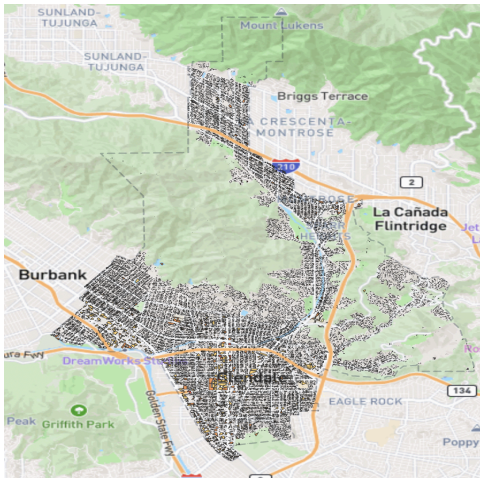


- Example of popup for clicked building

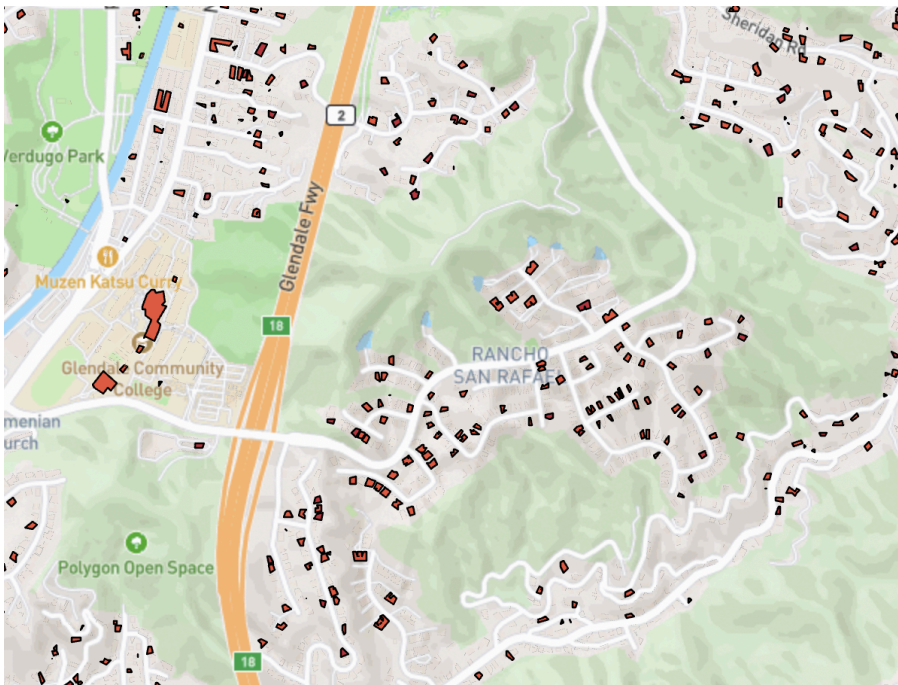
Attributes	
fid	51068
Height	26.433369657318497
Elevation	76.59079796848944
Fire Risk	2
Orbit	

2D Dashboard

- Leaflet top-down building risk map



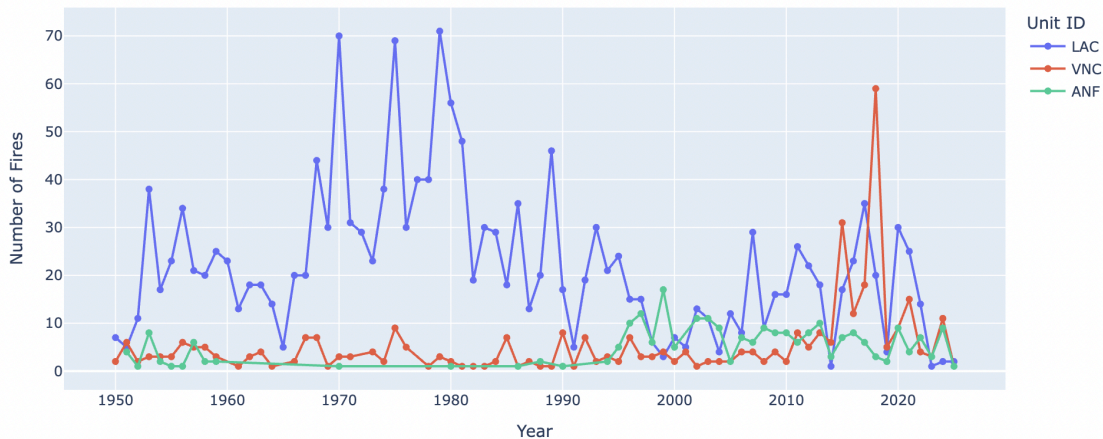
- Filter buttons (Snapshot of buildings in Rancho San Rafael neighborhood filtered to High-Risk)



Analysis Page (Supplemental Charts)

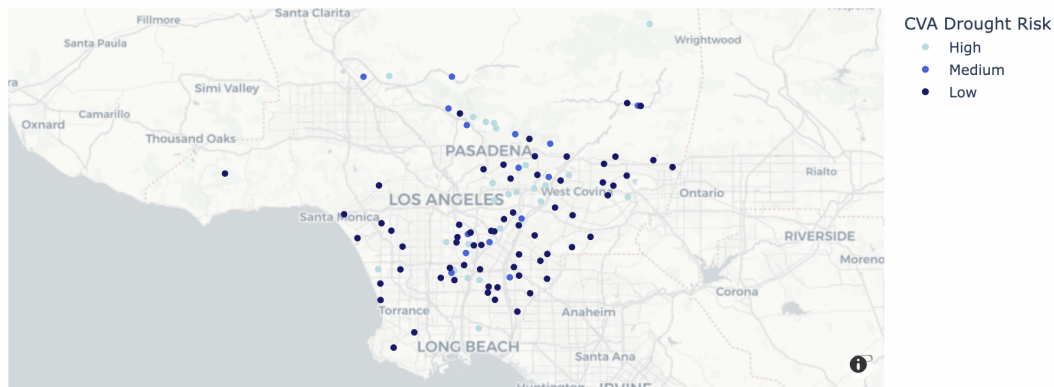
- Fire frequency chart

Historic Wildfire Counts Near Glendale



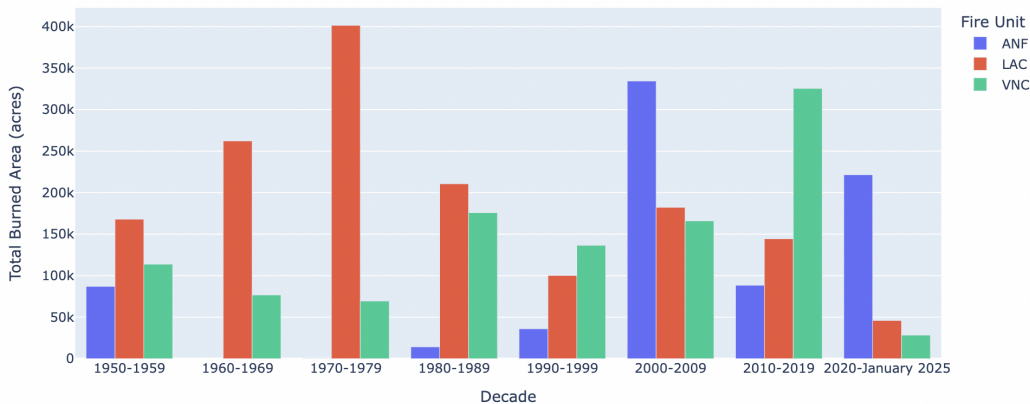
- Drought risk map

Los Angeles County Drought Risk Map



- Total acreage burned

Total Wildfire Burned Acres per Decade



B. Code Snippets

B.1 GIS Processing (QGIS)

- Reclassification (slope, aspect, vegetation)
 - Slope and aspect were reclassified via QGIS *Reclassify by Table*. (See main report for full tables.)
 - From the vegetation fire risk table in the main report, we reclassified each vegetation type into its risk score as follows:

$$("vegetation_glendale@1" = 154) * 1 + ("vegetation_glendale@1" = 664) * 1 +$$
$$("vegetation_glendale@1" = 696) * 3 + ("vegetation_glendale@1" = 747) * 2 +$$
$$("vegetation_glendale@1" = 845) * 2$$

- Over 100 vegetation IDs were reclassified

- Raster calculator expression for composite model:

$$(0.3 * "slope_risk@1") + (0.2 * "aspect_risk@1") + (0.5 * "veg_fire_risk@1")$$

B.2 Web Deployment

- Leaflet building-style and risk filter logic:

```
function showAll() {  
  buildingLayer.setStyle(f => {  
    fillColor: COLORS[f.properties["Fire Risk"]],  
    fillOpacity: 0.85,  
    weight: 0.4,  
    color: "#333" });}  
  
function filterRisk(minRisk) {  
  buildingLayer.setStyle(f => {  
    const r = f.properties["Fire Risk"];  
    return {fillColor: COLORS[r],  
      fillOpacity: (r >= minRisk) ? 0.9 : 0.05,  
      weight: (r >= minRisk) ? 1.2 : 0.1,  
      color: (r >= minRisk) ? "#000" : "#999" }; });}
```

- Snippet of Qgis2threejs initialization:

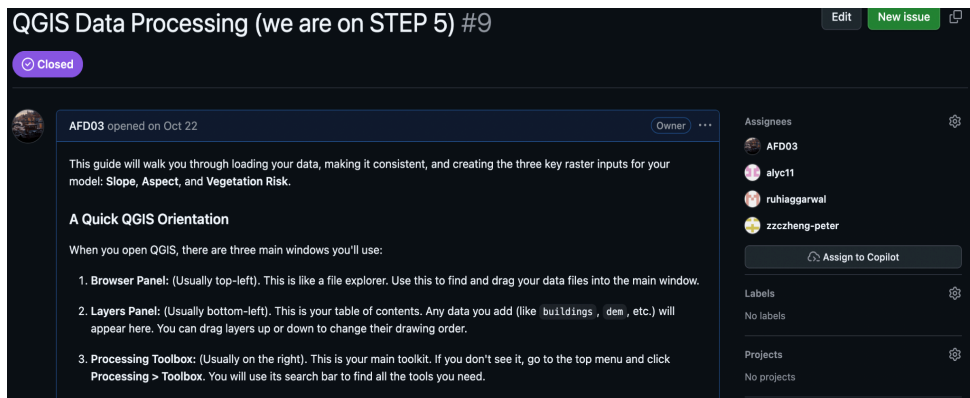
```
Var container = document.getElementById("view"),
app.init(container);

app.loadSceneFile("data/index/scene.json", function() {
    app.start();})
```

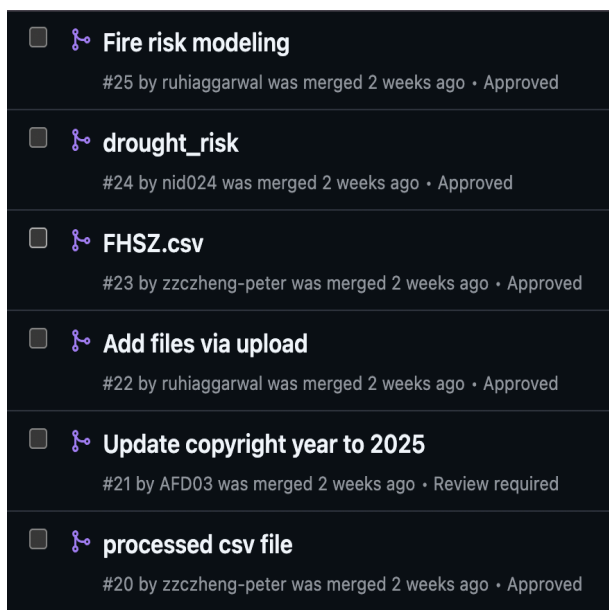
C. Collaboration

Examples:

- Tasks assigned via GitHub issues:



- Pull requests are used to add files and edit code in a clean and organized manner:



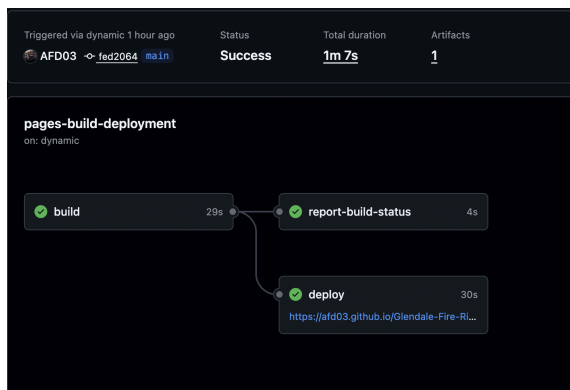
- GitHub Timeline:

Timeline

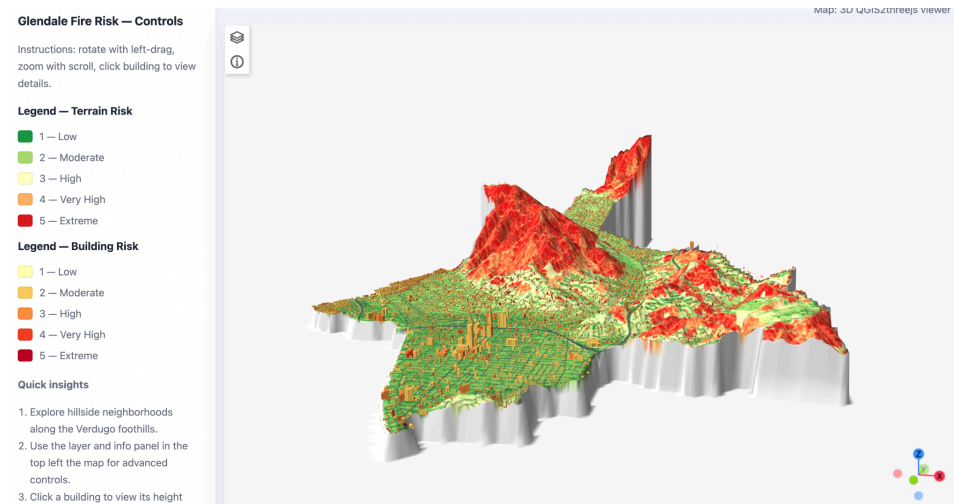
Week	Task
Week 4	Download & preprocess GIS data
Week 5	GIS analysis (slope, aspect, vegetation classification)
Week 6	Build and validate risk model
Week 7	Finalize GeoJSON + begin 2D visuals
Week 8	Complete 2D dashboard + 3D scene
Week 9	Integration + report writing
Week 10	Final testing & presentation

D. Deployment Evidence

- GitHub Pages Build



- Live map running in the browser



E. Workload Distribution & 480-Hour Effort Evidence

Our work spanned six weeks with structured weekly milestones. The estimated combined effort exceeded **480 hours**, consistent with STA160 expectations.

Weekly Timeline:

Week	Milestone	Estimated Hours
3	Data acquisition (DEM, WHR vegetation, building footprints, wildfire history), metadata review, CRS selection, and Glendale boundary preprocessing	55 hrs
4	Raster processing: slope & aspect generation, vegetation cleaning, NoData repair, clipping, resampling, raster alignment	65 hrs
5	Zonal statistics on buildings, centroid generation, height cleaning, attrib merge, CSV cleaning, EDA in Python	70 hrs
6	Statistical modeling: PCA, k-means clustering, elbow method tuning, rule-based fire-risk scoring, comparison of model vs logic results	75 hrs
7	Plotly + statistical visualizations: wildfire trends, drought/temperature plots, housing & elevation distributions, cluster diagnostic plots	65 hrs
8	Web deployment (Leaflet, risk filters, info popups, JS logic), 3D model (Qgis2threejs), animation testing, UI polishing, debugging	105 hrs
9	Report writing: methods, results, GIS workflow diagrams, reclassification tables, appendix construction, final proofing & editing	60 hrs